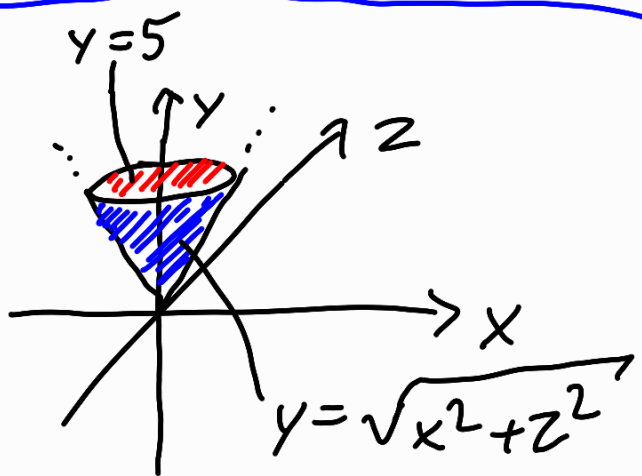


No announcements this week

Good luck on finals & whatever else

16: $\iint_S y^2 z^2 dS$, S is part of cone
 $y = \sqrt{x^2 + z^2}$ with $0 \leq y \leq 5$.

Step 1: Graph surface & find bounds



y bounds already given.
As for x & z , use
polar coordinates.

Entire surface fits
inside circle of radius

5 at the top of the cone. So let

$$x = r \cos \theta, \quad z = r \sin \theta$$

Then no restriction on θ , so $0 \leq \theta \leq 2\pi$.

$5 \geq y = \sqrt{x^2 + z^2} = r$ on the surface,

so $0 \leq r \leq 5$.

Step 2: Find ds & substitute

Let $f(x, z) = \sqrt{x^2 + z^2}$, then S is of the form $y = f(x, z)$. Recall that $ds = \sqrt{1 + f_x^2 + f_z^2} dx dz$ for such surfaces.

$$\text{So } \frac{ds}{dx dz} = \sqrt{1 + \left(\frac{\partial x}{\partial \sqrt{\dots}}\right)^2 + \left(\frac{\partial z}{\partial \sqrt{\dots}}\right)^2} = \sqrt{1 + \frac{x^2}{x^2 + z^2} + \frac{z^2}{x^2 + z^2}} = \sqrt{1 + \frac{x^2 + z^2}{x^2 + z^2}} = \sqrt{2}.$$

Step 3: Substitute bounds & evaluate.

$$\iint_S y^2 z^2 ds = \iint_S (x^2 + z^2) z^2 \sqrt{2} dx dz.$$

$$dx dz = r dr d\theta, \quad x^2 + z^2 = r^2, \quad z^2 = r^2 \sin^2 \theta$$

Note: There is no dy since this is a surface integral and not a volume integral.

So integrand is $r^5 \sin^2 \theta \sqrt{2} dr d\theta$,

so new integral is

$$\begin{aligned}
 & \int_0^5 \int_0^{2\pi} r^5 \sin^2 \theta \sqrt{2} \, dr \, d\theta = \\
 & \sqrt{2} \left(\int_0^5 r^5 \right) \left(\int_0^{2\pi} \frac{1 - \cos 2\theta}{2} \, d\theta \right) = \\
 & \sqrt{2} \left. \frac{r^6}{6} \right|_0^5 \left(\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right) \Big|_0^{2\pi} = \\
 & \sqrt{2} \cdot \frac{5^6}{6} \cdot \pi = \frac{15625\pi\sqrt{2}}{6},
 \end{aligned}$$

noting that $5^6 = 125^2 = 15625$.

22: $\iint_S \vec{F} \cdot d\vec{S}$, $\vec{F} = \langle x, y, z \rangle$, S
 param. by $\vec{r}(u, v) = \langle u, v, u^2 + v^2 \rangle$
 for $0 \leq u, v \leq 1$ & oriented upward

Step 1: Find dS & substitute
 You can use $d\vec{S} = (\vec{r}_u \times \vec{r}_v) \, dA$,
 but it's easier to note that

S is just the surface $z=f(x,y)$
for $f(x,y) = x^2 + y^2$. So

$$dS = \langle -f_x, -f_y, 1 \rangle dA =$$
$$\underline{\langle -2x, -2y, 1 \rangle dx dy}.$$

And $u=x, v=y$, so bounds
are $0 \leq x, y \leq 1$.

Step 2: Substitute F, dS &
simplify integrand

$$F = \langle x, y, z \rangle = \langle x, y, x^2 + y^2 \rangle,$$

so $F \cdot dS = \langle x, y, z \rangle \cdot \langle -2x, -2y, 1 \rangle$

$$= (-2x^2 - 2y^2 + z) dx dy$$
$$= -(x^2 + y^2) dx dy.$$

Step 3: Evaluate integral

$$\iint_S F \cdot dS = - \int_0^1 \int_0^1 (x^2 + y^2) dx dy$$

$$= - \int_0^1 \left(\frac{x^3}{3} + y^2 x \right) \Big|_0^1 dy =$$

$$- \int_0^1 \left(\frac{1}{3} + y^2 \right) dy = - \left(\frac{1}{3} + \frac{1}{3} \right) = -\frac{2}{3}$$

3b: $F = \langle x e^y, z - e^y, -xy \rangle$, S is $x^2 + 2y^2 + 3z^2 = 4$, find $\iint_S F \cdot d\vec{S}$.

Step 1: Find $\operatorname{div} F$.

$$\operatorname{div} F = (x e^y)_x + (z - e^y)_y + (-xy)_z$$

$$= e^y - e^y + 0 = 0$$

Step 2: Apply Theorem.

Let E enclose S . Then

$$\iint_S F \cdot d\vec{S} = \iiint_E \operatorname{div} F \, dV = \iiint_E 0 \, dV$$

$$= 0$$

12: $\iint_S x^2 y z \, dS$, $z = 1 + 2x + 3y$ above $[0, 3] \times [0, 2]$.

Step 1: Find dS

Let $f(x, y) = 1 + 2x + 3y$. Then S is described by $z = f(x, y)$, so

$$dS = \sqrt{1 + f_x^2 + f_y^2} \, dx \, dy = \sqrt{1 + 2^2 + 3^2} \, dx \, dy = \underline{\sqrt{14} \, dx \, dy}$$

Step 2: Substitute bounds, dS , eval

$$x^2 y z \, dS = x^2 y (1 + 2x + 3y) \sqrt{14} \, dx \, dy,$$

so the integral is

$$\begin{aligned} & \int_0^2 \int_0^3 x^2 y (1 + 2x + 3y) \sqrt{14} \, dx \, dy = \\ & \sqrt{14} \int_0^2 y \int_0^3 (x^2 + 2x^3 + 3yx^2) \, dx \, dy \\ & = \sqrt{14} \int_0^2 y \left[\frac{x^3}{3} + \frac{x^4}{2} + yx^3 \right] \Big|_0^3 \, dy \\ & = \sqrt{14} \int_0^2 y \left(9 + \frac{81}{2} + 27y \right) \, dy \end{aligned}$$

$$= \sqrt{14} \int_0^2 \left(\frac{99}{2} y + 27y^2 \right) dy =$$

$$\sqrt{14} \left(\frac{99}{4} y^2 + 9y^3 \right) \Big|_0^2 =$$

$$\sqrt{14} (99 + 72) = 171\sqrt{14}.$$

1c: $\iint_S z \, dS$, $r(u,v) = \langle u^2, v, u \rangle$
 (u,v) ranges over $[0,1] \times [0,1]$.

Step 1: Redescribe S .

Notice $u = z$, $v = y$, so $x = u^2 = z^2$

for $0 \leq u, v \leq 1 \implies 0 \leq y, z \leq 1$.

Let $f(y,z) = z^2$, then S is
 the surface $x = f(y,z)$.

Step 2: Find $z \, dS$

$$dS = \sqrt{1 + f_y^2 + f_z^2} \, dy \, dz =$$

$$\sqrt{1 + 0 + (2z)^2} \, dy \, dz = \sqrt{4z^2 + 1} \, dy \, dz,$$

$$\text{so } z \, dS = \underline{z(4z^2 + 1)^{1/2} \, dy \, dz}$$

Step 3: Plug in & evaluate:

$$\begin{aligned}\iint_S z \, dS &= \int_0^1 \int_0^1 z(4z^2+1)^{1/2} \, dy \, dz = \\&= \int_0^1 z(4z^2+1)^{1/2} \, dz = \frac{2}{3} \frac{1}{8} (4z^2+1)^{3/2} \Big|_0^1 \\&= \frac{1}{12} (5^{3/2} - 1^{3/2}) = \frac{5\sqrt{5} - 1}{12}.\end{aligned}$$
